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Academic Year 2025 – 26

S5 PROJECT WORK

Final Review

PROJECT TITLE

HEALTHCARE MANAGEMENT SYSTEM USING OPENEMR

BIP PROJECT ID

27S5MIN234

CATEGORY

EXTERNAL – OWN

BATCH MEMBERS

7376232CB121 – HARIHARAN K

7376232CB154 – SUBHA R

7376232CB126 – KAVIPRIYA P

7376231CS338 – UDHAYATHARUN R

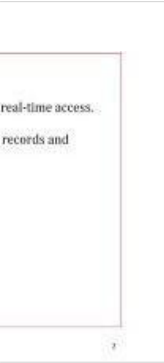
GUIDE

SARANYA N

Assistant Professor Level-I

Department of Computer Science And Business Systems

Problem Statement



PROBLEM STATEMENT

Traditional healthcare record-keeping is inefficient, error-prone, and lacks real-time access.

There's a growing need for a scalable and secure system to manage patient records and hospital operations digitally.

ABSTRACT

ABSTRACT :

This project aims to build a fully functional healthcare management system using OpenEMR, a widely adopted open-source electronic health record (EHR) and medical practice management solution.

The system supports core healthcare operations such as patient registration, appointment scheduling, medical billing, prescriptions, and reporting. The project is implemented locally using XAMPP , APACHE and MySQL, making it suitable for clinics, hospitals, and research purposes.

AIM & OBJECTIVES OF THE PROJECT

AIM :

To implement and deploy an open-source healthcare management facilitates efficient and secure handling of patient and medical data.

OBJECTIVES :

- *Provide a centralized digital platform for hospitals and clinics.*
- *Enable secure Electronic Medical Records (EMR) storage and access.*
- *Automate appointments, billing, and patient monitoring.*
- *Ensure user-friendly access for both doctors and staff.*

Literature Survey/Problem Identification and Setup

Sl.No.	Journal Paper Title with Author	Works carried out (with details of Methods/ Materials/ Software/ Algorithms / fabrication / techniques/ components used)	Information gathered relevant to your project
1	Shanmugam, P., & Kumar, R. (2022). Open-source EHR solutions in small-scale healthcare practices. Int. Journal of Medical Informatics, 58(3), 101-110. https://journals.sagepub.com/doi/10.1177/14604582221099828	Analyzed the use of OpenEMR for managing electronic health records and clinical workflows in small healthcare environments.	Demonstrated how OpenEMR simplifies hospital operations and offers cost-effective digital healthcare solutions.
2	Abdalkareem, Z. A., et al. (2021). *Healthcare scheduling in optimization context: a review*. **Health and Technology**, 11, 445–469. https://link.springer.com/article/10.1007/s12553-021-00547-5	Reviewed **190 research studies** on healthcare scheduling. Methods included **metaheuristics (GA, PSO), (MILP), hybrid methods**, and simulation-based scheduling.	Guides designing an **appointment & resource scheduling module** in your web system to reduce waiting time and improve doctor/room utilization.

Literature Survey/Problem Identification and Setup

Sl.No.	Journal Paper Title with Author	Works carried out (with details of Methods/ Materials/ Software/ Algorithms / fabrication / techniques/ components used)	Information gathered relevant to your project
3	Abdelrahman Mahmoud, & Jason J. Saleem. (2024). *Usability Assessment of OpenEMR: Enhancing Healthcare Interactions*. SAGE Journals. https://www.researchgate.net/publication/387190432_Usability_Assessment_of_OpenEMR_Enhancing_Healthcare_Interactions	Evaluated **OpenEMR usability** using SUS and heuristic methods, analyzing interface efficiency, navigation, and error handling.	Helps refine **front-end modules** (patient registration, vitals entry, user access) and improve **UI/UX** for smoother integration with backend ML features.
4	Ramanna, S., Pinnamraju, J., Rallabandi, S. S., Ayyam, V. K., Edara, Y. R. K., Malempati, T., Pilli, M., Moncy, M. M., Ratna, S., Somasundaram, M., Kokkalakonda, M., Hoyt, R., & faculty. (2025). *Lessons Learned from OpenEMR Implementation*. PubMed Central. https://pubmed.ncbi.nlm.nih.gov/40417551/	Described **hands-on implementation of OpenEMR** in an academic setting. Covered installation, customization, and training for healthcare informatics students.	. Provides practical lessons for **deploying OpenEMR**, highlighting customization, training, and adoption challenges relevant to your project.

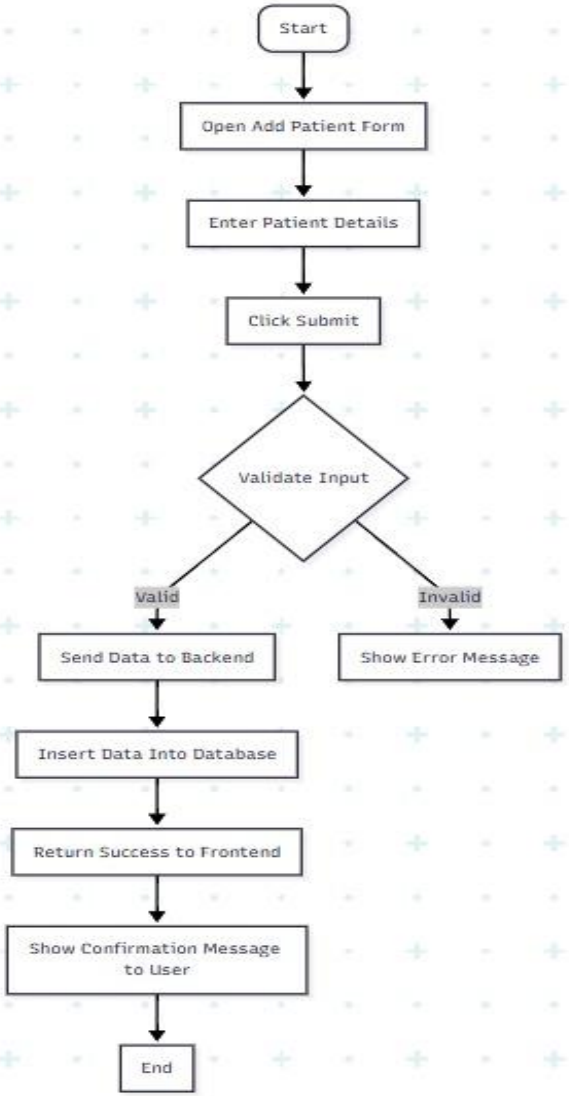
Literature Survey/Problem Identification and Setup

Sl.No	Journal Paper Title with Author	Works carried out (with details of Methods/ Materials/ Software/ Algorithms / fabrication / techniques/ components used)	Information gathered relevant to your project
5	Purkayastha, S. et al. (2019). *Open-Source EHR Comparison: OpenEMR Case*. Healthc Inform Res. https://scholarworks.indianapolis.iu.edu/server/api/core/bitstreams/71aa2d4b-56f8-490a-8191-219ed198381f/content	Used qualitative criteria based on IOM requirements and Meaningful Use to compare five OSS EHRs; performance measured via task-time benchmarks.	Highlights that **OpenEMR meets all 32 functional criteria**—the strongest among peers—while **OpenMRS offers better task performance**, offering valuable context as you design both front-end and back-end (ML) modules.
6	Noll, J., Beecham, S., & Seichter, D. (2011). *Qualitative Study of OpenEMR Development*. ESEM 2011.https://pure.ul.ie/en/publications/a-qualitative-study-of-open-source-software-development-the-opene	Analysed OpenEMR forum discussions using content analysis and inter-rater reliability (Cohen's kappa). Found that core contributors include clinicians and that a small expert group can sustain the project with high responsiveness	demonstrates the importance of involving domain experts (e.g., clinicians) during development and shows how a focused, responsive core team can effectively support specialized software projects—valuable insight as you integrate backend ML features.

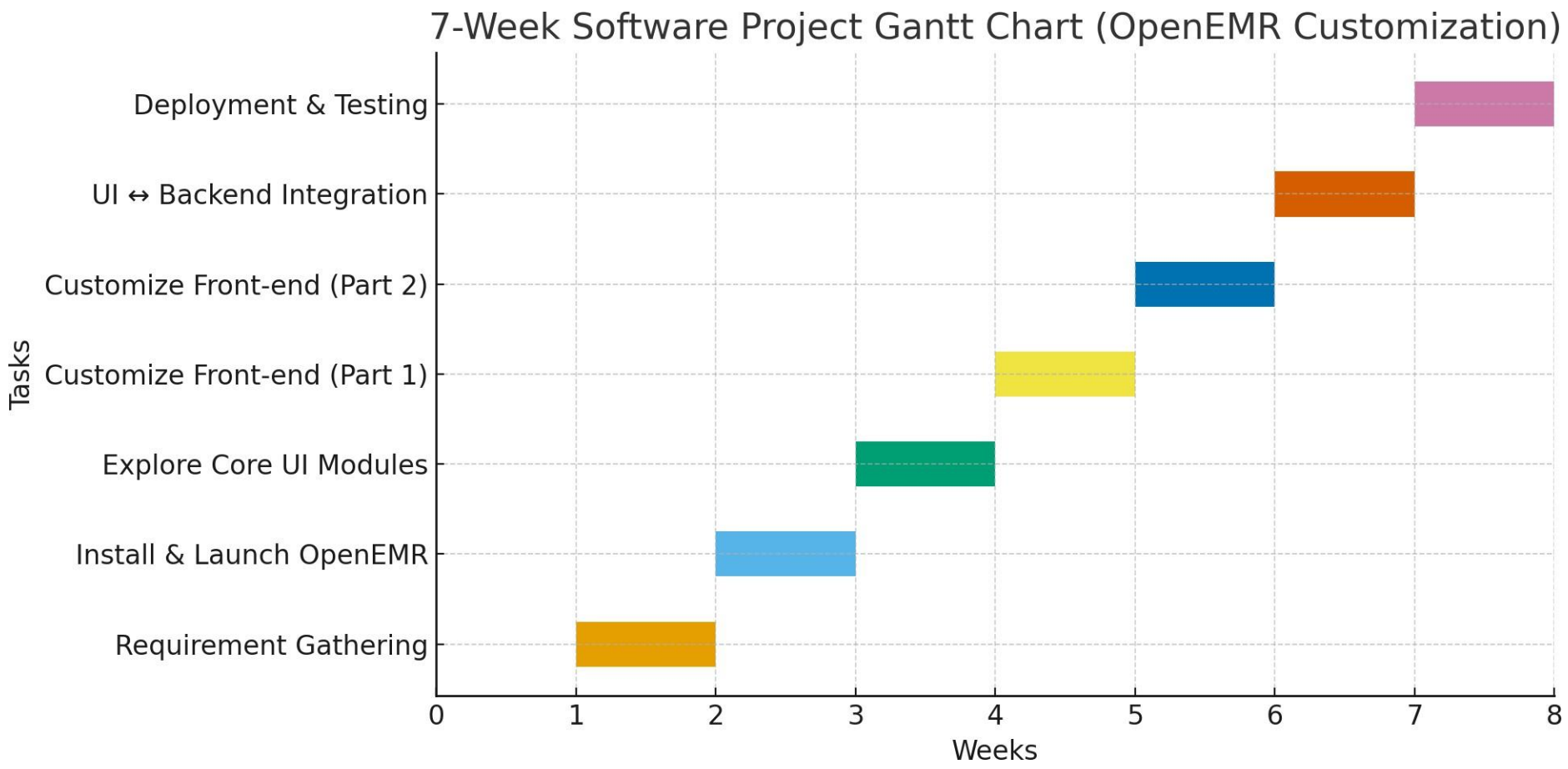
Literature Survey/Problem Identification and Setup

Sl.No	Journal Paper Title with Author	Works carried out (with details of Methods/ Materials/ Software/ Algorithms / fabrication / techniques/ components used)	Information gathered relevant to your project
7	Kessy, E. C., Kibusi, S. M., & Ntwenya, J. E. (2024). *EMR Data Use in Decision-Making*. Frontiers in Digital Health. https://pubmed.ncbi.nlm.nih.gov/38414827/	Cross-sectional survey of 308 health managers in Tanzanian primary care; logistic regression to identify predictors of EMR data use: access to computers, EMR coverage across work areas, information quality perception, and data literacy.	Highlights that **EMR data use is driven more by infrastructure, coverage, and perceived quality than training alone**. This is directly relevant to your project—especially in designing backend systems and workflows that ensure reliable data access and quality to support decision-making and ML-driven anomaly detection.
8	Muinga, N. et al. (2018). *Open-Source EHR in Kenyan Public Facilities*. JMIR Med Inform. https://pmc.ncbi.nlm.nih.gov/articles/PMC5932328/	Case study via interviews and stakeholder visits; implementation of a modified OpenMRS (AfyaEHMS), infrastructure upgrades, user training, local–international collaboration; initial deployment scaled back due to sociotechnical and administrative challenges	Highlights real-world barriers: need for **user buy-in**, **local support**, phased rollout, infrastructure readiness, and adaptability to local context—all crucial for your front-end and backend design, especially in ML integration and deployment strategy.

PROPOSED METHODOLOGY (Flow Chart)



PROPOSED METHODOLOGY (Gantt Chart)



PROPOSED WORK AND METHODOLOGY

This project implements a Healthcare Management System using OpenEMR to automate key hospital functions like patient registration, appointments, billing, prescriptions, and medical records. It features role-based access control for data security and is tailored for small to medium-sized clinics, leveraging open-source tools to minimize costs.

METHODOLOGY :

1. *Requirement Analysis:Gather functional requirements for doctors, patients, and administrative staff.*
2. *Technology Selection:Select OpenEMR as the base healthcare management platform due to its modular design, open-source availability, and compliance with healthcare standards.Use MySQL database for data storage and PHP/Apache server for backend hosting.*
3. *System Design:Customize OpenEMR modules according to hospital needs.*
4. *Implementation*
5. *Testing:Perform module-wise testing*
6. *Deployment and Demonstration:*
7. *Deploy the system in a test environment (local host) for demonstration purposes.*
8. *Documentation & Reporting: Prepare user manuals and technical documentation and report project outcomes*

NEED FOR THE CURRENT STUDY

- Healthcare organizations face challenges in managing patient records, scheduling, and hospital workflows efficiently. Although many proprietary Electronic Health Record (EHR) systems exist, their high cost and limited customization restrict their adoption in small- and medium-scale healthcare facilities. OpenEMR, being open-source, offers a cost-effective solution but requires significant customization and integration to meet modern healthcare demands.
- In the current study, the need arises from:

- **Efficient Digitalization of Healthcare Records** → To ensure accurate and accessible patient data management.
- **Customization of OpenEMR Front-End** → To improve user experience for hospital staff through better UI, forms, and dashboards.
- **Backend Integration** → To enable smooth data flow, secure storage, and system reliability.
- **Resource Scheduling & Workflow Automation** → To reduce waiting time, optimize doctor/room utilization, and improve service delivery.

SCOPE OF THE PROJECT

- *The Healthcare Management System using OpenEMR aims to develop a centralized, secure, and open-source platform for managing healthcare operations digitally.*
- *This system will streamline hospital workflows by automating patient registration, appointment scheduling, billing, and electronic health records (EHR) management.*
- *Implementation of OpenEMR for managing medical records, appointments, and billing.*
- *Secure Role-Based Access Control (RBAC) for doctors, patients, and administrative staff.*
- *Deployment of the system in a local server environment (XAMPP) or containerized environment (Docker) for flexibility.*
- *Integration of modules like Patient registration & management, Appointment booking system, Prescription generation, Billing & payment tracking, Lab test management, Reporting & analytics for healthcare data.*
- *User-friendly web interface for easy access by doctors, staff, and patients.*

INDIVIDUAL CONTRIBUTIONS TO THE WORK

Batch Member 1 : (7376232CB121 – HARI HARAN K)

•Developed the **frontend** of the Healthcare Management System, including user interface design, dashboards, forms, and interactive elements.

•Implemented the **backend** functionality, including server-side scripting, business logic, API development, and integration with the database.

INDIVIDUAL CONTRIBUTIONS TO THE WORK

Batch Member 2 : (7376232CB154 – SUBHA R)

•Focused on **backend** development, implementing server-side logic, API endpoints, and integration with OpenEMR.

•Contributed to **database (DB)** management, handling CRUD operations, data integrity, and synchronization with frontend modules.

INDIVIDUAL CONTRIBUTIONS TO THE WORK

Batch Member 3 : (7376232CB126 – KAVIPRIYA P)

- Developed portions of the **frontend**, including UI components, forms, and validation.

- Assisted in **database (DB)** design and maintenance, ensuring proper relationships, indexing, and efficient data handling.

INDIVIDUAL CONTRIBUTIONS TO THE WORK

Batch Member 4 : (7376231CS338 – UDHAYATHARUN R)

•Worked on the **frontend**, designing responsive web pages, layouts, and interactive dashboards.

•Contributed to the **database (DB)** implementation, including table design, relationships, and query optimization.

THANK YOU